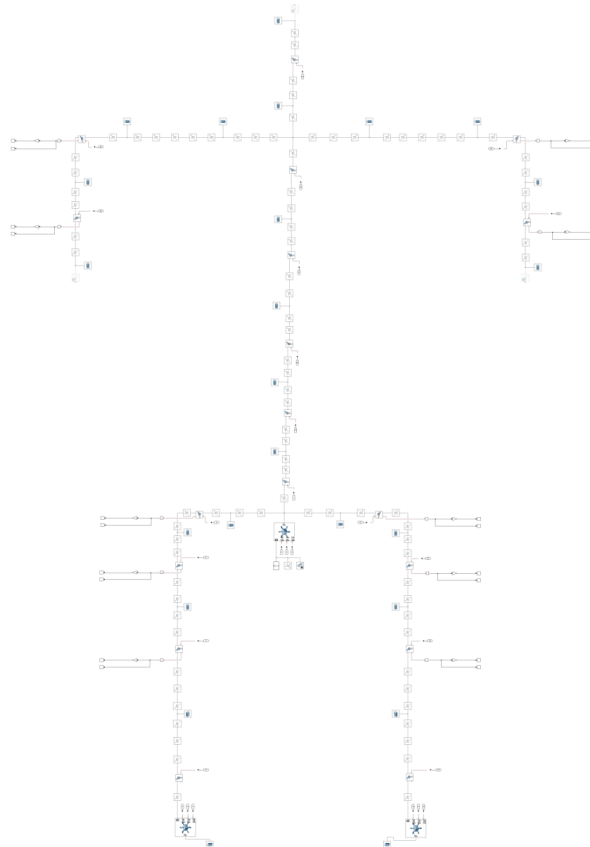


Subject-Specific Sagittal Human Motion Reconstruction

and Validation in MATLAB/Simscape

Computational biomechanics project



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Motion-capture processing, subject-specific parameterization, multibody reconstruction, and quantitative validation

Table of Contents

Abstract.....	2
1. Objective and application context.....	2
2. Input data and coordinate system	2
3. Processing pipeline	3
3.1 Joint-center extraction	3
3.2 N-pose reference	3
3.3 Sagittal segment inclinations	3
3.4 Relative joint rotations.....	3
4. Multibody model	3
5. Visualization	4
6. Validation methodology	5
6.1 Joint-center reconstruction	5
6.2 Quaternion-based angle cross-validation	8
6.3 Parent-child link-vector decomposition	12
7. Results and interpretation.....	16
7.1 Joint-angle validation.....	16
7.2 Spinal chain reconstruction.....	16
7.3 T8-to-shoulder-root geometry.....	16
7.4 Shoulder-to-upper-arm offsets.....	16
7.5 Upper-arm-to-forearm offsets.....	16
7.6 Forearm-to-hand connections	16
7.7 Accumulation of the right-hand position error.....	17
7.8 Lower-limb chain.....	17
7.9 Fixed bias components.....	17
7.10 Dynamic residuals	19
8. Technical diagnosis	19
9. Scope of the conclusions.....	19
10. Conclusion	20

Abstract

This project implements an end-to-end workflow for reconstructing measured human postural motion in MATLAB and Simscape Multibody. A full-body motion-capture recording containing 23 anatomical segments is converted into joint-center trajectories, a subject-specific N-pose reference, sagittal segment inclinations, and relative joint rotations. The calculated motion drives a multibody human model whose segment lengths are parameterized from the measured geometry.

A dedicated validation layer compares reconstructed sagittal joint-center coordinates with the original motion-capture coordinates, cross-checks position-derived angles against sagittal angles derived from segment-orientation quaternions, and decomposes parent-child link-vector errors into fixed geometric bias and time-varying residual components. The results show close agreement in joint-angle trajectories and sub-millimeter reconstruction discrepancies along the spinal chain. The largest upper-limb position differences are primarily caused by fixed anatomical frame and link-placement offsets rather than dynamic angle errors.

1. Objective and application context

The aim is to reconstruct postural motion recorded during an Achilles-tendon vibration experiment. The selected interval covers 45-65 s of the measurement, including the first vibration period between 50 and 60 s. The reconstruction provides two complementary visualizations:

- a subject-scaled Simscape Multibody representation driven by calculated sagittal joint rotations,
- a direct stick-figure animation based on measured joint-center coordinates.

The project focuses on the engineering pipeline from measurement data to a transparent and customizable multibody representation.

2. Input data and coordinate system

The input is a MATLAB structure converted from a motion-capture recording.

- Sampling frequency: 100 Hz
- Number of body segments: 23
- Analysis interval: frames 4500:6500
- Global coordinate convention:
 - +x: forward,
 - +y: leftward,
 - +z: upward.

For each frame, the motion-capture structure provides:

- global proximal joint-center positions,
- segment-orientation quaternions,
- segment and joint labels.

The dataset used in this project was anonymized by removing the original participant filename, recording path, and recording date.

3. Processing pipeline

3.1 Joint-center extraction

The main MATLAB workflow reads the ordered motion-capture segment labels and assigns each triplet of global position values to the corresponding body segment. This produces a structured set of trajectories for the pelvis, spinal levels, upper limbs, lower limbs, feet, and toe roots.

3.2 N-pose reference

The Simscape model is constructed in an N-pose. The source recording contains a T-pose reference, so the arm coordinates in the reference frame are rearranged to form a geometrically consistent N-pose while preserving measured arm-segment lengths.

3.3 Sagittal segment inclinations

For a segment vector projected into the sagittal plane,

$$\mathbf{v}_{xz}(t) = \begin{bmatrix} v_x(t) \\ v_z(t) \end{bmatrix},$$

the global segment inclination is calculated with:

$$\phi(t) = \text{atan2}(v_z(t), v_x(t)).$$

The rotation relative to the N-pose is:

$$\theta(t) = -[\phi(t) - \phi_N].$$

The negative sign follows the rotation convention used by the Simscape model.

3.4 Relative joint rotations

The model is a serial multibody chain. Relative rotations are therefore obtained by subtracting the orientation of the proximal segment from the orientation of the distal segment. Examples include:

$$\theta_{knee} = \theta_{lower\ leg} - \theta_{upper\ leg},$$

$$\theta_{elbow} = \theta_{forearm} - \theta_{upper\ arm},$$

$$\theta_{L3} = \theta_{L3\ segment} - \theta_{L5\ segment}.$$

These relative rotations are applied to revolute joints in the Simscape model.

4. Multibody model

The Simscape Multibody model represents the major segments of the spine, arms, pelvis, legs, feet, and toe roots. Segment geometry is parameterized from the motion-capture reference configuration. The pelvis translation and sagittal joint rotations are prescribed from the measurement-derived signals.

The model is intended as a kinematic motion-reconstruction environment. It is not used here to estimate validated joint torques, muscle forces, or predictive balance-control responses.

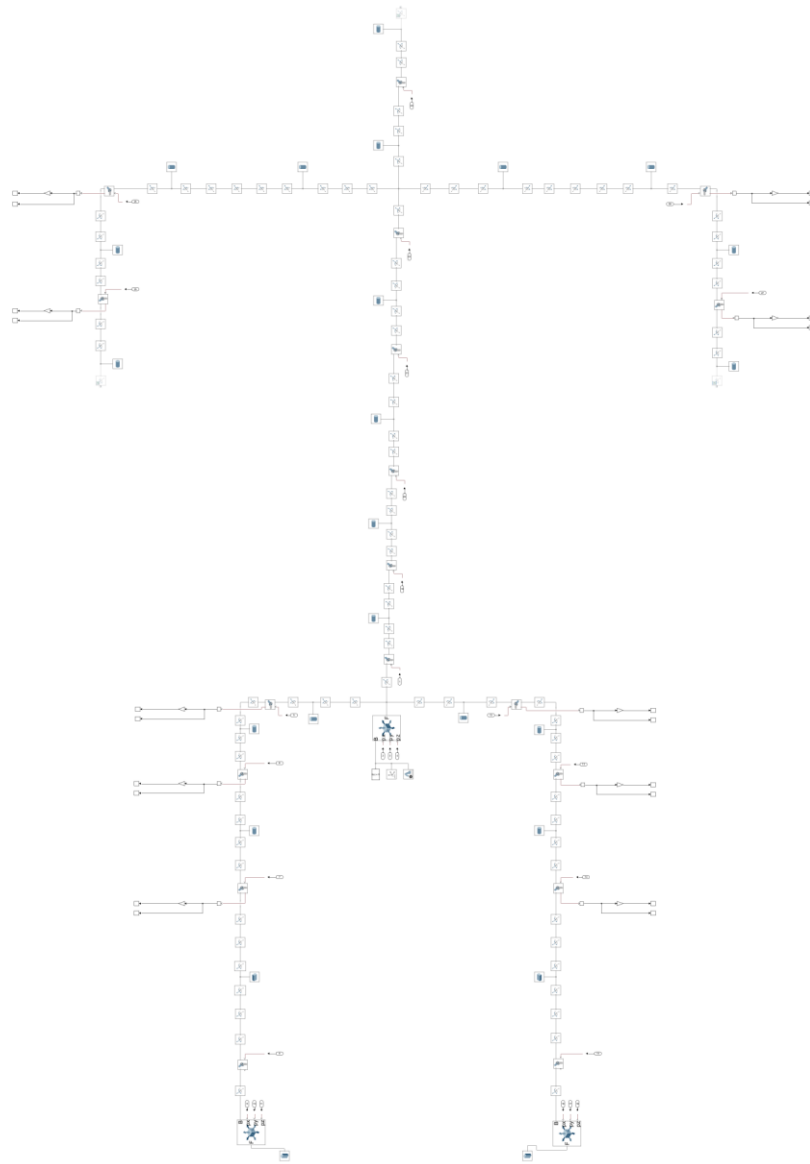


Figure 1. Overview of the subject-scaled Simscape Multibody model architecture.

5. Visualization

The measured motion is visualized in two forms:

1. A Simscape Multibody animation showing the subject-scaled mechanical representation.
2. A three-dimensional stick figure assembled directly from motion-capture joint-center coordinates and viewed in the sagittal plane.

The two views provide a direct comparison between measured geometry and the multibody reconstruction.

6. Validation methodology

6.1 Joint-center reconstruction

The Simscape model is solved kinematically for every frame. Model joint-center coordinates are recovered in the sagittal plane and compared with the measured motion-capture coordinates.

For joint center i :

$$\mathbf{e}_i(t) = \mathbf{p}_{i,model}(t) - \mathbf{p}_{i,Mocap}(t),$$

$$e_i(t) = \sqrt{e_{i,x}^2(t) + e_{i,z}^2(t)}.$$

The reported metrics are mean error, maximum error, and RMSE:

$$RMSE_i = \sqrt{\frac{1}{N} \sum_{t=1}^N e_i^2(t)}.$$

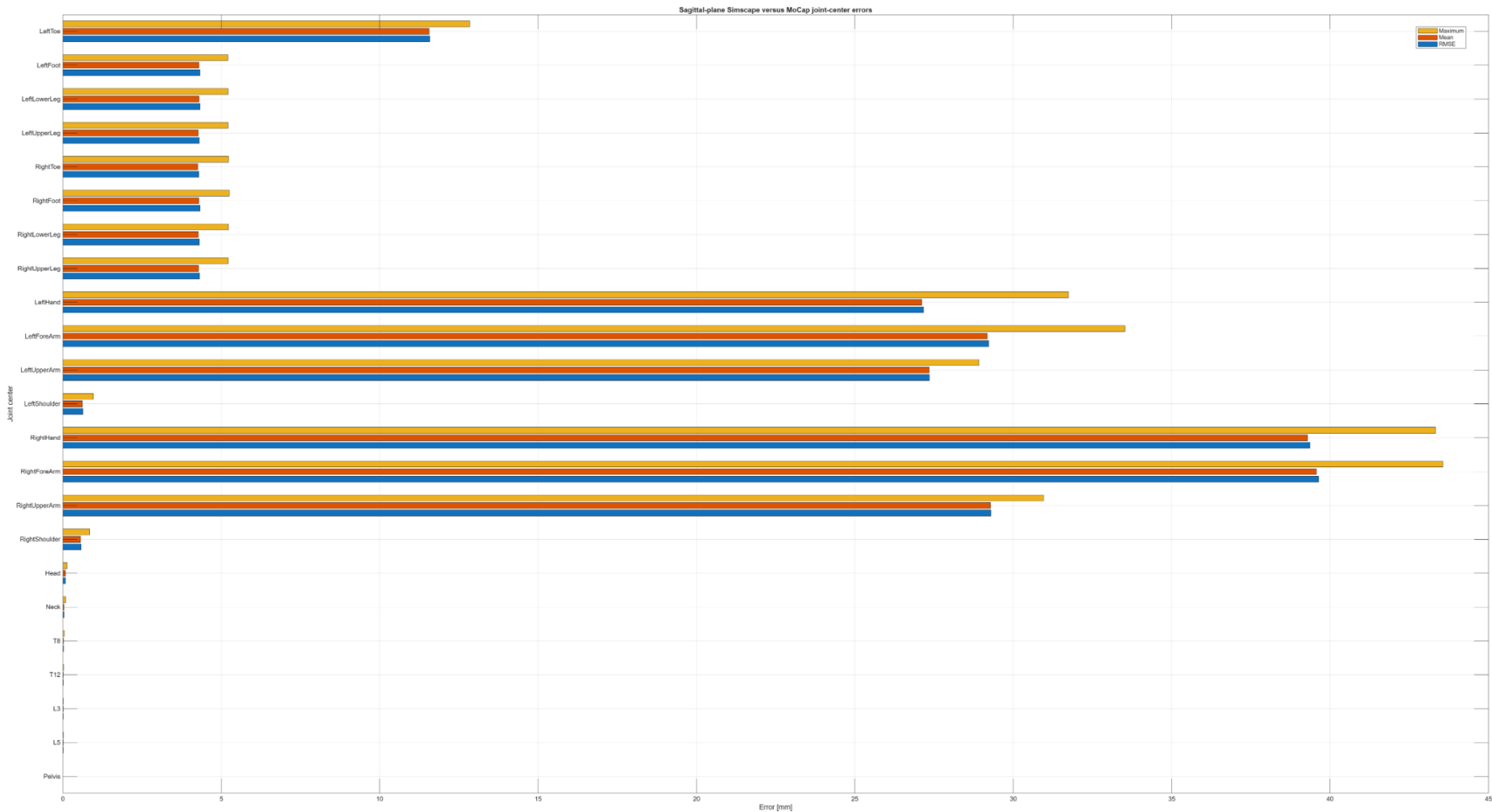


Figure 2. Sagittal-plane joint-center error summary, including maximum, mean, and RMSE values.

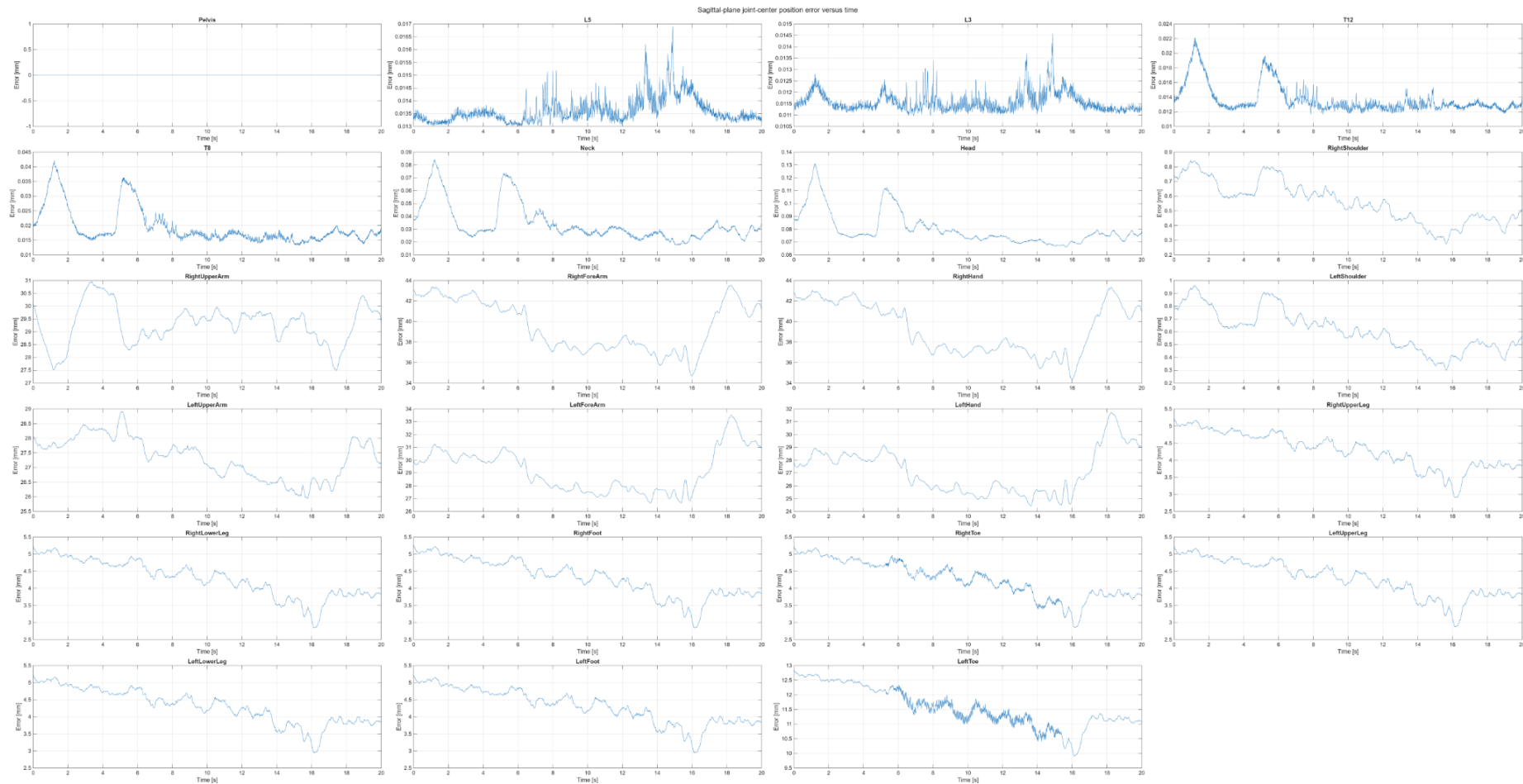


Figure 3. Sagittal-plane joint-center position error over time for each anatomical point.

6.2 Quaternion-based angle cross-validation

A second sagittal angle estimate is derived from motion-capture segment-orientation quaternions. The segment-local anatomical axis is identified from the original T-pose position vector and T-pose quaternion. The axis is then rotated by the quaternion of each measured frame and referenced to the N-pose direction.

The position-derived and quaternion-derived angle trajectories are compared through:

- RMSE,
- mean absolute error,
- maximum absolute error,
- correlation,
- time-series overlays.

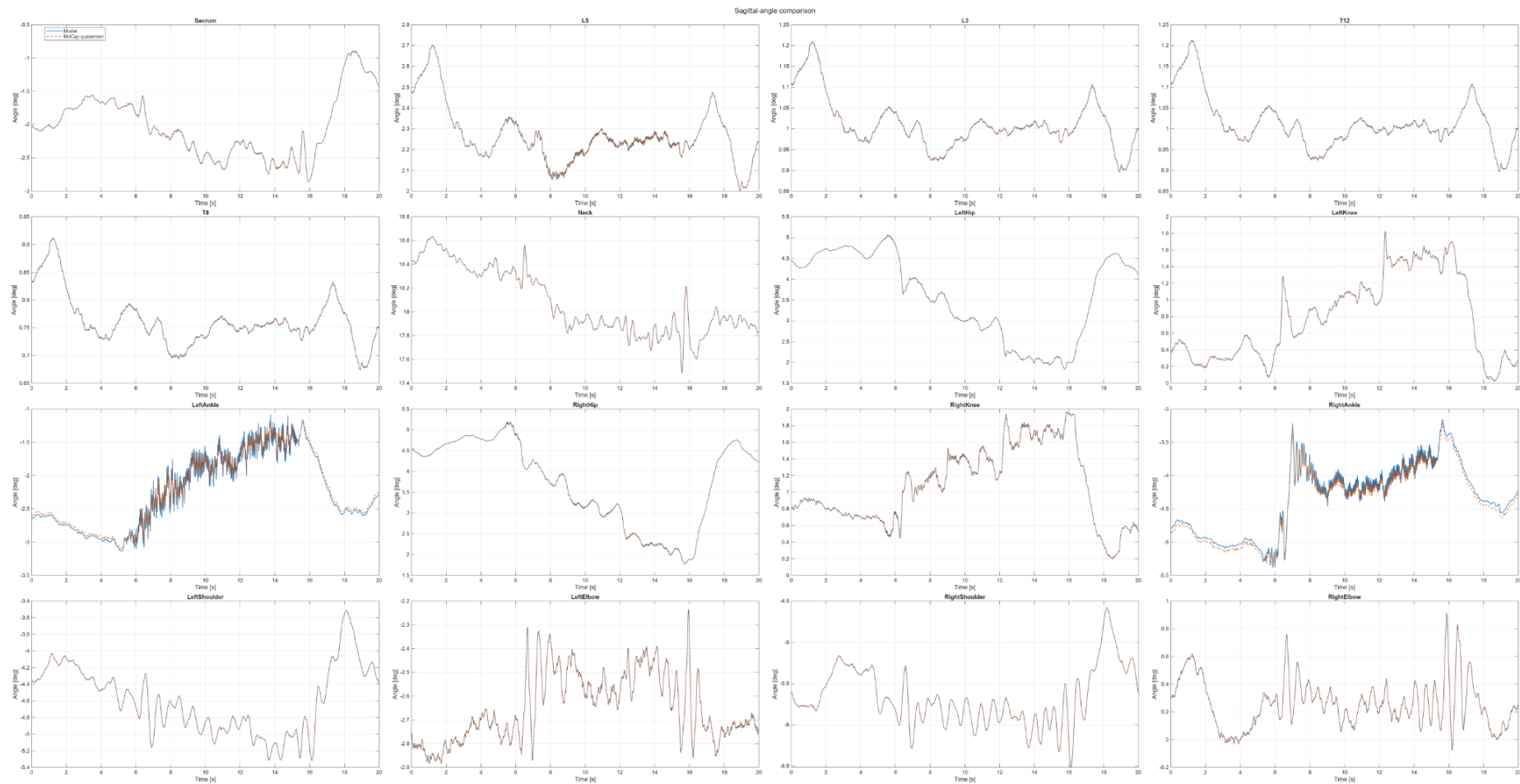


Figure 4. Comparison of position-derived model angles and quaternion-derived sagittal angles.

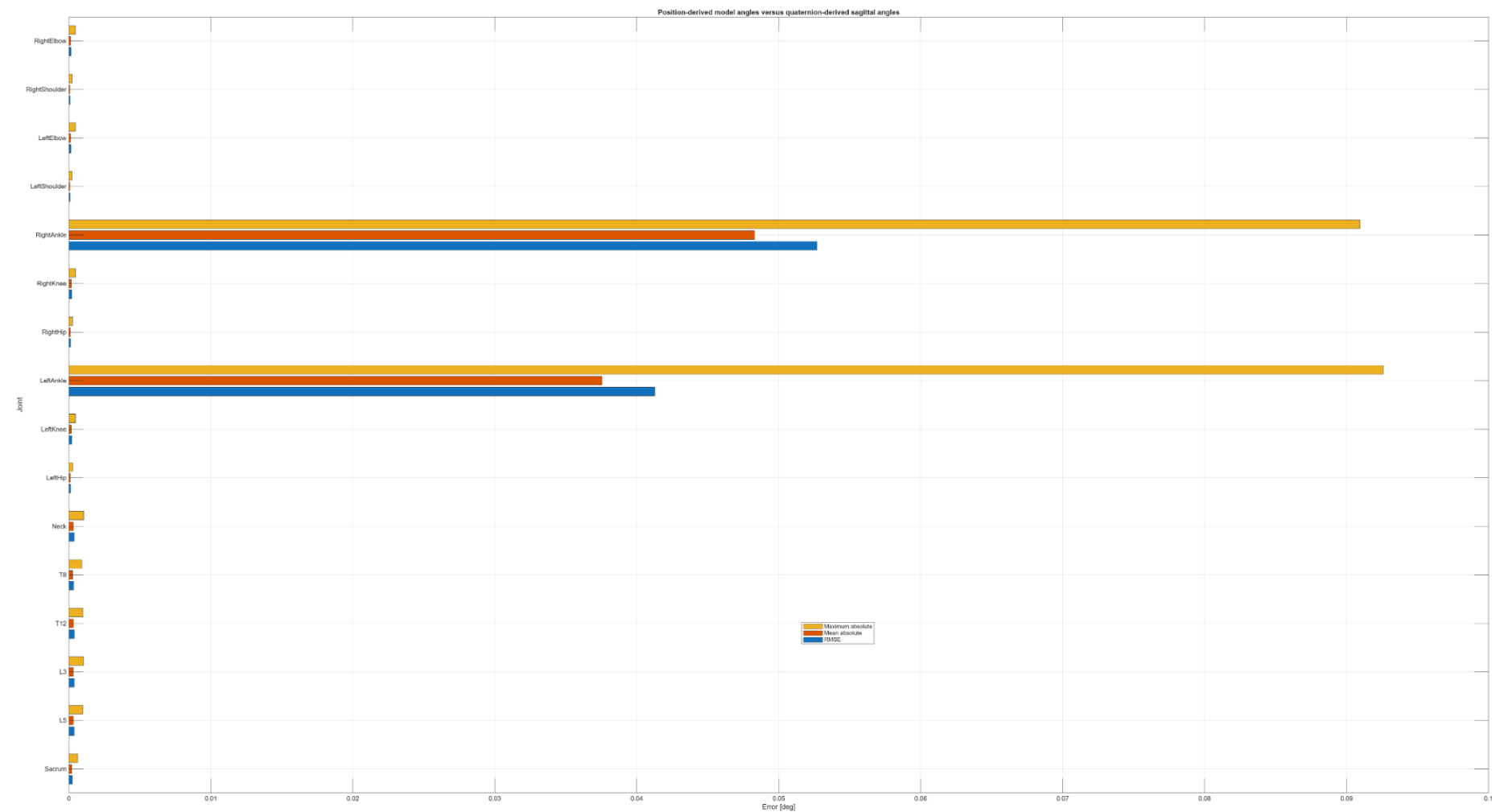


Figure 5. Summary of angular validation metrics: maximum absolute error, mean absolute error, and RMSE.

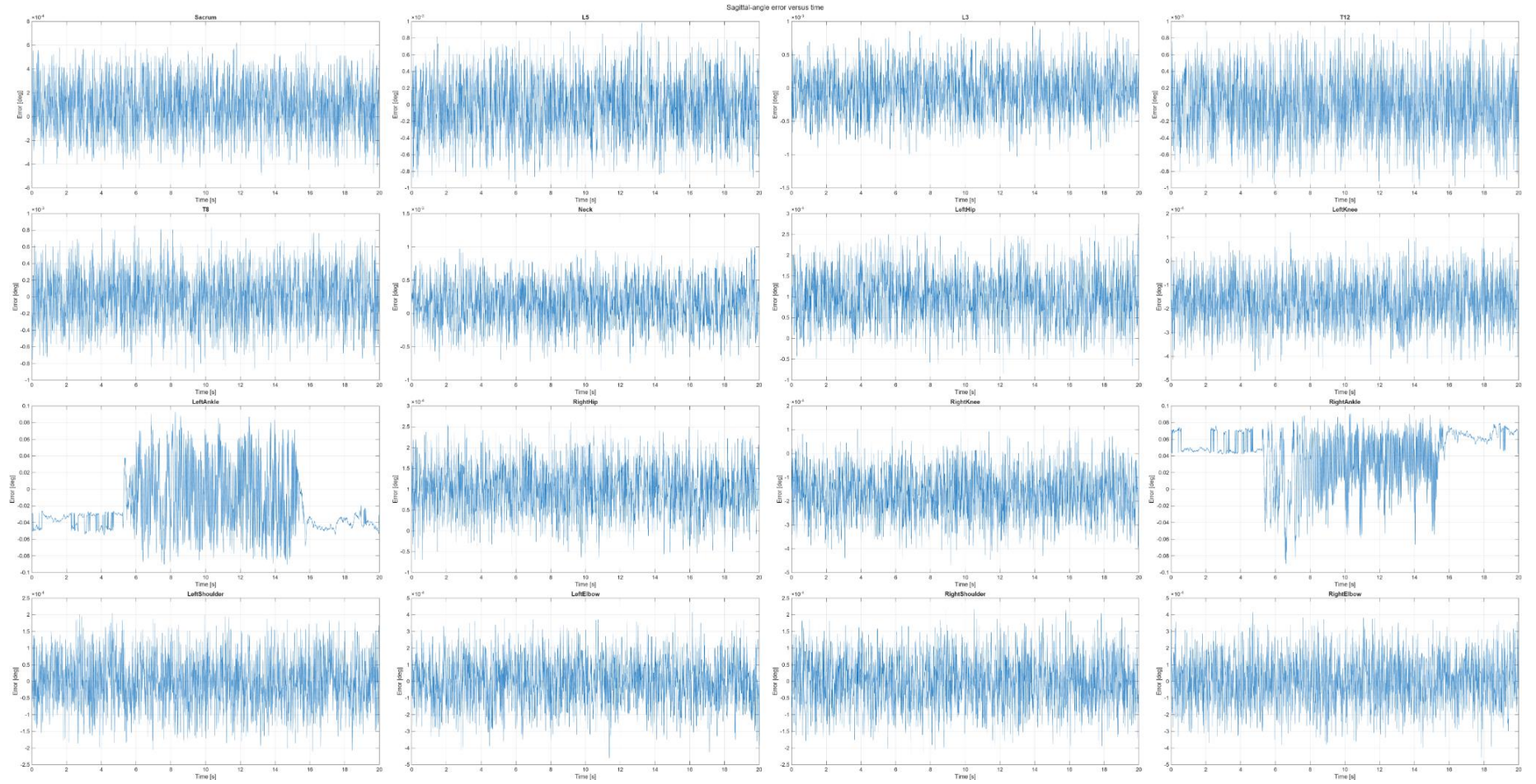


Figure 6. Sagittal-angle error over time for the evaluated joints.

6.3 Parent-child link-vector decomposition

Absolute joint-center errors can accumulate across a serial chain. To localize their source, each parent-child vector is compared directly:

$$\begin{aligned}\mathbf{r}_{model}(t) &= \mathbf{p}_{child,model}(t) - \mathbf{p}_{parent,model}(t), \\ \mathbf{r}_{MoCap}(t) &= \mathbf{p}_{child,MoCap}(t) - \mathbf{p}_{parent,MoCap}(t), \\ \Delta \mathbf{r}(t) &= \mathbf{r}_{model}(t) - \mathbf{r}_{MoCap}(t).\end{aligned}$$

The mean link-vector difference is treated as a fixed geometric bias:

$$\overline{\Delta \mathbf{r}} = \frac{1}{N} \sum_{t=1}^N \Delta \mathbf{r}(t).$$

The dynamic residual is:

$$\Delta \mathbf{r}_{dynamic}(t) = \Delta \mathbf{r}(t) - \overline{\Delta \mathbf{r}}.$$

This decomposition distinguishes static landmark and frame-placement differences from motion-dependent kinematic errors.

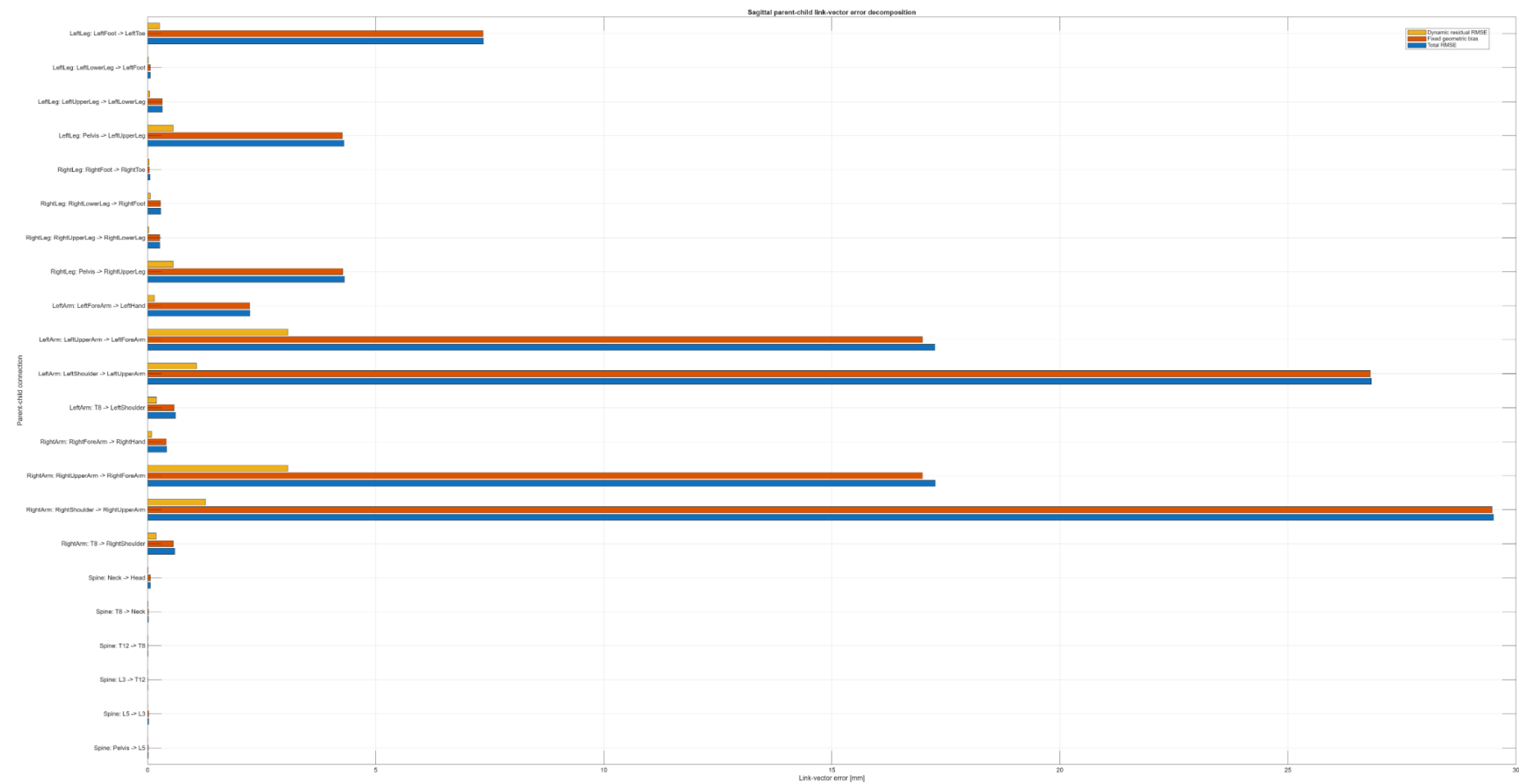


Figure 7. Parent-child link-vector error decomposition into total RMSE, fixed geometric bias, and dynamic residual RMSE.

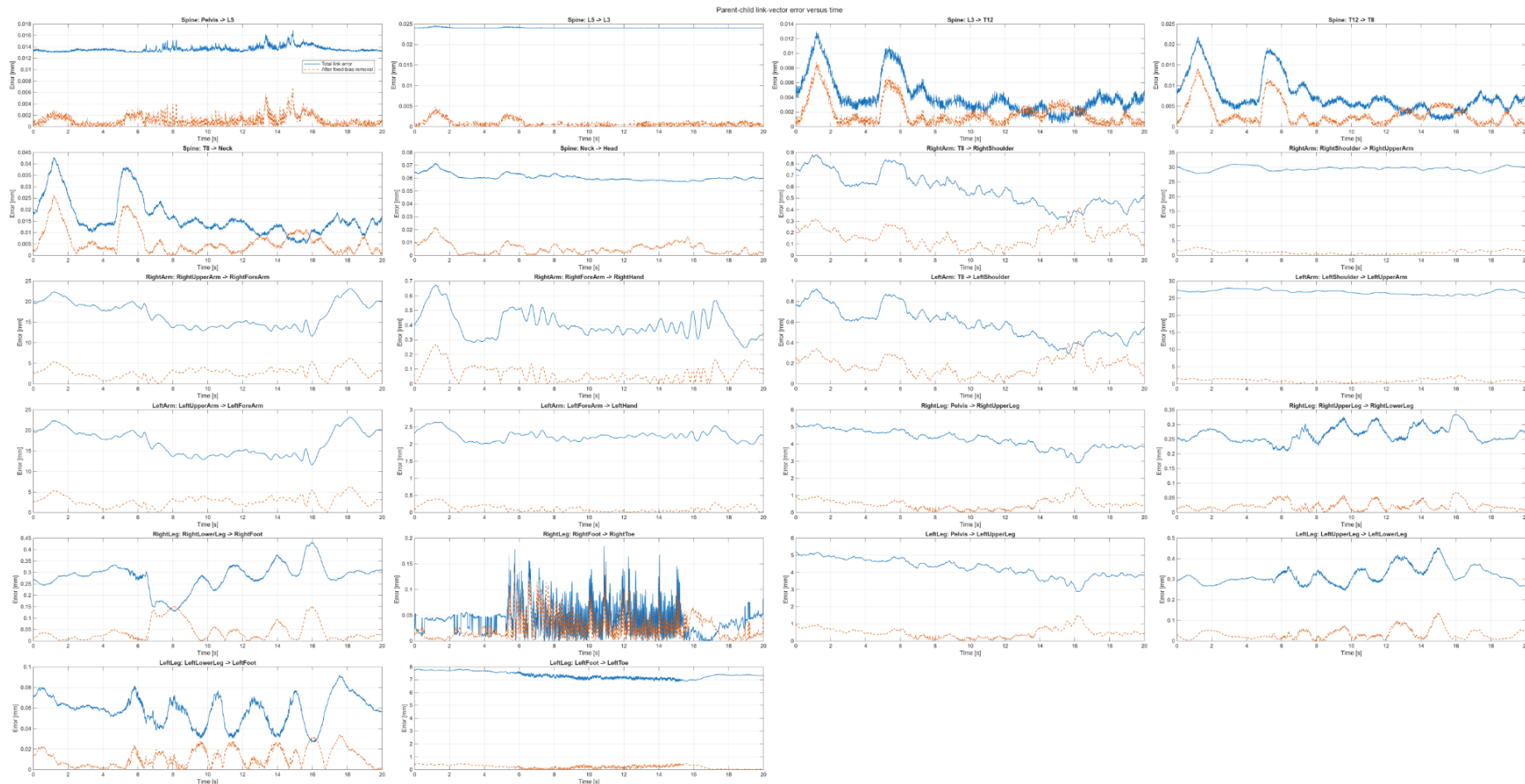


Figure 8. Parent-child link-vector error over time before and after removal of the fixed geometric bias.

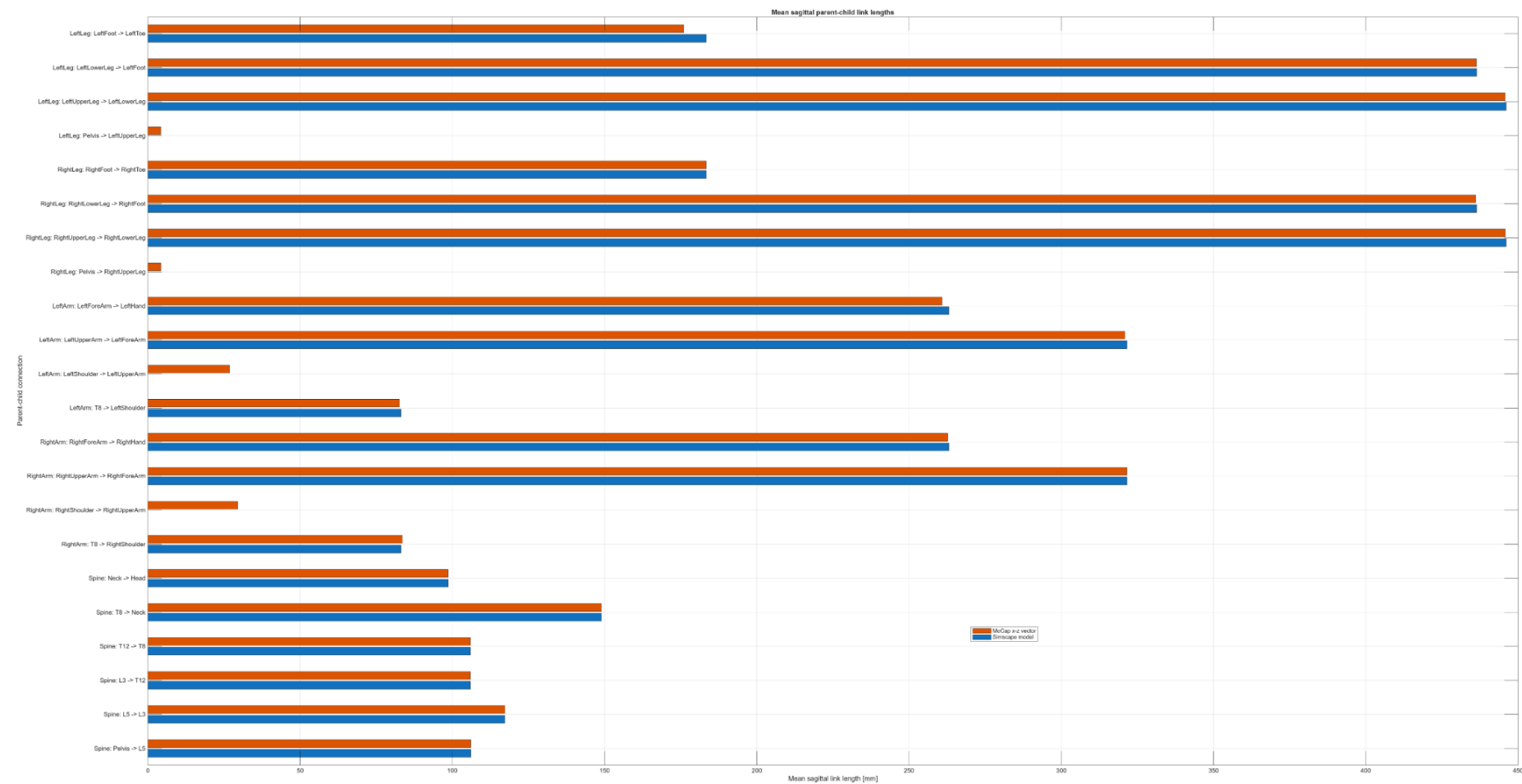


Figure 9. Mean sagittal parent-child link lengths in the motion-capture data and the Simscape model.

7. Results and interpretation

7.1 Joint-angle validation

The position-derived and quaternion-derived sagittal angle trajectories overlap closely across the spine, hips, knees, shoulders, and elbows. For most non-ankle joints, the differences are at numerical-noise scale. The ankle comparisons show the largest angular differences, but remain below approximately 0.1 deg.

This agreement supports the internal consistency of the sagittal-angle extraction and relative-joint formulation with the motion-capture segment-orientation data.

7.2 Spinal chain reconstruction

The pelvis-to-head chain shows very small joint-center and link-vector discrepancies. The pelvis is a prescribed translational reference, while the L5, L3, T12, T8, neck, and head trajectories are reconstructed with sub-millimeter numerical error.

This indicates consistent segment geometry, joint topology, and relative-angle transfer through the spinal chain.

7.3 T8-to-shoulder-root geometry

The T8-to-left-shoulder-root and T8-to-right-shoulder-root links remain below approximately one millimeter of error. This indicates that the upper-thorax-to-scapula-root mapping is consistent after matching the model and motion-capture anatomical points.

7.4 Shoulder-to-upper-arm offsets

The dominant upper-limb discrepancy begins at the shoulder-root-to-upper-arm connection:

- right side: approximately 29 mm,
- left side: approximately 27 mm.

The link decomposition shows that nearly all of this difference is fixed geometric bias, while the dynamic residual is small. The Simscape shoulder-root and shoulder-joint frames therefore differ from the corresponding motion-capture landmark geometry by an almost constant sagittal offset.

7.5 Upper-arm-to-forearm offsets

The upper-arm-to-forearm connections show an additional error of approximately 17 mm on both sides. The mean sagittal link lengths remain similar, indicating that the discrepancy is not primarily caused by segment-length mismatch.

The dominant fixed bias is consistent with a small constant difference in the link's zero orientation or local frame alignment. For an upper-arm length of roughly 320 mm, a 17 mm endpoint difference corresponds to an angular offset on the order of three degrees.

7.6 Forearm-to-hand connections

The right forearm-to-hand link is reconstructed closely, with a sub-millimeter error. The left side contains a smaller fixed difference of approximately 2 mm. These results indicate that wrist placement contributes little to the right-hand error and only a limited amount to the left-hand error.

7.7 Accumulation of the right-hand position error

The absolute right-hand difference approaches approximately 40 mm. This is not produced by one isolated 40 mm error. It results from vector accumulation of:

1. the shoulder-root-to-upper-arm fixed offset,
2. the upper-arm-to-forearm fixed orientation/frame offset,
3. the much smaller forearm-to-hand difference.

Because these components have different directions, their magnitudes do not add arithmetically. Their vector combination produces the observed distal hand error.

7.8 Lower-limb chain

The main lower-limb difference appears in the pelvis-to-hip-joint connection and is approximately 4-5 mm on both sides. The upper-leg-to-lower-leg and lower-leg-to-foot connections remain below approximately 0.5 mm.

The lower-limb angles and long-segment geometry are therefore reconstructed consistently. The dominant difference is a small fixed pelvis-to-hip landmark offset rather than a time-varying joint-angle error.

The left foot-to-toe connection contains an additional fixed difference of approximately 7 mm, explaining why the left-toe absolute error is higher than the remaining lower-limb points.

7.9 Fixed bias components

The fixed-bias components show the direction of each geometric difference.

- The shoulder-to-upper-arm links contain a dominant vertical component and, on the right side, an additional anterior-posterior component.
- The upper-arm-to-forearm links contain a predominantly anterior-posterior bias.
- The pelvis-to-hip offsets are small and largely constant.
- The left foot-to-toe link contains a distinct fixed length and direction difference.

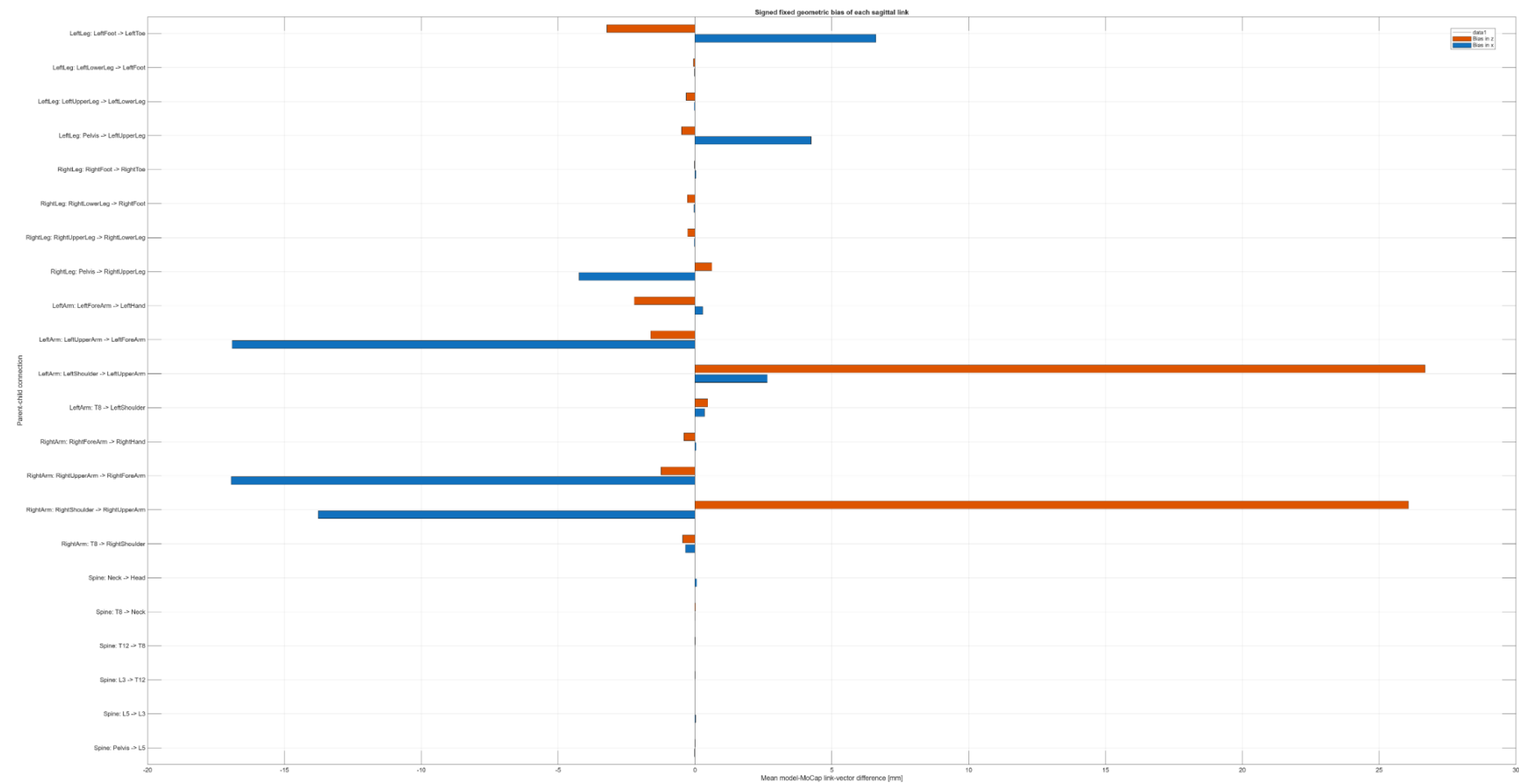


Figure 10. Signed fixed geometric bias components for each sagittal parent-child link.

7.10 Dynamic residuals

For the links with the largest absolute errors, the dynamic residual is much smaller than the fixed bias:

- shoulder-to-upper-arm: roughly 1 mm dynamic residual versus 27-30 mm fixed bias,
- upper-arm-to-forearm: roughly 3 mm dynamic residual versus approximately 17 mm fixed bias,
- pelvis-to-hip: sub-millimeter dynamic residual versus approximately 4-5 mm fixed bias,
- left foot-to-toe: sub-millimeter dynamic residual versus approximately 7 mm fixed bias.

This indicates that the model generally follows the temporal motion pattern correctly. Most remaining joint-center differences arise from fixed geometry and landmark definitions.

8. Technical diagnosis

The validation supports the following diagnosis:

Components verified as consistent

- sagittal angle extraction from position vectors,
- quaternion-based angle cross-validation,
- relative-joint-angle calculation,
- spinal-chain topology,
- hip-knee-ankle kinematic transfer,
- upper-thorax-to-shoulder-root mapping,
- temporal tracking of measured postural motion.

Remaining fixed geometric differences

- shoulder root to shoulder joint,
- upper-arm local zero orientation or frame alignment,
- pelvis center to hip-joint placement,
- left foot to toe-root placement,
- a smaller left forearm-to-hand difference.

9. Scope of the conclusions

The project demonstrates a transparent and quantitatively evaluated kinematic reconstruction workflow. It does not claim:

- independent experimental validation against optical motion capture,
- predictive simulation of postural responses,
- validated inverse dynamics,
- muscle-force estimation,
- full three-dimensional inverse kinematics.

The angle comparison is an internal cross-validation between motion-capture position and orientation outputs. The joint-center analysis evaluates how the measured kinematics are represented by the sagittal multibody geometry.

10. Conclusion

The model accurately reconstructs sagittal joint-angle trajectories and reproduces the spinal and lower-limb serial chains with small dynamic residuals. The dominant upper-limb and selected distal-limb position differences are attributable to fixed anatomical frame and link-placement offsets rather than time-varying kinematic errors.

The resulting project is a complete computational biomechanics case study that combines motion-capture processing, subject-specific parameterization, multibody modeling, visualization, quantitative validation, and structured error diagnosis.

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